Full-waveform inversion Least-squares migration

Ketil Hokstad, February 2016
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Non-linear inversion recap

Objective function:

$$\Phi = \frac{1}{2}[\mathbf{d} - \mathcal{F}(\mathbf{m})]^T \mathbf{\Sigma}_n^{-1} [\mathbf{d} - \mathcal{F}(\mathbf{m})] + \frac{1}{2}[\mathbf{m} - \mathbf{m}_0]^T \mathbf{\Sigma}_m^{-1} [\mathbf{m} - \mathbf{m}_0]$$

Jacobian matrix:

$$\mathbf{J} = \frac{\partial \mathcal{F}}{\partial \mathbf{m}}$$

Dimensions of the Jacobi matrix: $N_d \times N_m$

Then

$$\mathbf{g} = \mathbf{J}^H \mathbf{\Sigma}_n^{-1} \Delta \mathbf{d} + \frac{\partial C}{\partial \mathbf{m}}$$

$$\mathbf{H} \simeq \mathbf{J}^H \mathbf{\Sigma}_n^{-1} \mathbf{J} + \frac{\partial^2 C}{\partial \mathbf{m}^2}$$

Gauss-Newton

(neglecting $\partial^2 \mathcal{F} / \partial \mathbf{m}^2$ terms)

- Matrix free: Use adjoint-state (back-propagation) methods
- Approximations to inverse Hessian: BFGS, L-BFGS

Elastic Full-Waveform Inversion (FWI)

Elastic objective function

$$\begin{aligned}\Phi &= \frac{1}{2}\int dt \int dS_r[\rho(\mathbf{x}_r)\Delta v_i(\mathbf{x}_r,t)\Delta v_i(\mathbf{x}_r,t) + c_{ijkl}\Delta\epsilon_{ij}(\mathbf{x}_r,t)\Delta\epsilon_{kl}(\mathbf{x}_r,t)] \\ &= \text{kinetic} + \text{potential "misfit energy"}\end{aligned}$$

Alternative form of 2nd term

$$c_{ijkl}\Delta\epsilon_{ij}\Delta\epsilon_{kl} = s_{ijkl}\Delta\sigma_{ij}\Delta\sigma_{kl} \quad (s_{ijkl} = \text{compliance})$$

Isotropic case;

$$c_{ijkl} = \delta_{ij}\delta_{kl}\lambda + (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\mu$$

Lame parameters:

$$\begin{aligned}v_P^2 &= \frac{\lambda + 2\mu}{\rho} \\ v_S^2 &= \frac{\mu}{\rho}\end{aligned}$$

Throw-back to our simple EM toy model

Consider EM, isotropic case.
Objective function in frequency domain:

$$\Phi = \frac{1}{4\pi} \int d\omega \int_R d^2R |\Delta E(\mathbf{x}_r, \omega)|^2 + \text{regularization}$$

$$g(\mathbf{x}) = \frac{-1}{2\pi} \int d\omega \int d^2R \, [K^*(\mathbf{x}_r, \omega, \sigma(\mathbf{x}))\Delta E(\mathbf{x}_r, \omega)]$$

$$K = \frac{\partial \mathcal{F}}{\partial \sigma} = \text{Frechet kernel}$$

Born approximation:

$$G(\mathbf{x}_r, \mathbf{x}_s) = G_0(\mathbf{x}_s, \mathbf{x}_r) + i\omega\mu_0 \int d^3x \, G_0(\mathbf{x}, \mathbf{x}_s)\delta\sigma(\mathbf{x})G_0(\mathbf{x}, \mathbf{x}_r)$$

- Single-scattering approximation

Frechet derivative

$$\begin{aligned}\delta G(\mathbf{x}_r,\mathbf{x}_s) &= i\omega\mu_0 \int d^3x\, G(\mathbf{x},\mathbf{x}_s)\delta\sigma(\mathbf{x})G_0(\mathbf{x},\mathbf{x}_r)\\ \frac{\delta G}{\delta\sigma} &= G_0(\mathbf{x},\mathbf{x}_s)G_0(\mathbf{x},\mathbf{x}_r)\end{aligned}$$

Account for the dipole source current

$$\begin{aligned}K(\mathbf{x}_r,\mathbf{x}) &= i\omega\mu_0 G_0(\mathbf{x}_r,\mathbf{x})\int d^3x_s G_0(\mathbf{x},\mathbf{x}_s)J(\mathbf{x}_s)\\ &= i\omega\mu_0 G_0(\mathbf{x}_r,\mathbf{x})E_0(\mathbf{x})\end{aligned}$$

Substitute for K^* in equation for g

$$g(\mathbf{x}) = \frac{1}{2\pi}\int d\omega\, i\omega\mu_0 E_0^*(\mathbf{x}) \underbrace{\int d^2R\, G_0^*(\mathbf{x},\mathbf{x}_r)\Delta E(\mathbf{x}_r)}_{\text{Back-propagation}}$$

Back to seismic FWI …

Frechet derivative

$$\delta\Phi = \frac{1}{2}\int\int\int d^3x[g_\rho(\mathbf{x})\delta\rho(\mathbf{x}) + g_\lambda(\mathbf{x})\delta\lambda(\mathbf{x}) + g_\mu(\mathbf{x})\delta\mu(\mathbf{x})]$$

(same exercise with Lippmann-Schwinger and Born approximation)

Gradients, isotropic:

$$g_\rho(\mathbf{x}) = -\int_0^T dt[\partial_t v_i^{FW}\partial_t v_i^{BW}] \quad \text{(sum over } i\text{)}$$

$$g_\lambda(\mathbf{x}) = \int_0^T dt[\partial_t \epsilon_{ii}^{FW}\partial_t \epsilon_{jj}^{BW}] \quad \text{(sum over } i, j\text{)}$$

$$g_\mu(\mathbf{x}) = 2\int_0^T dt[\partial_t \epsilon_{ij}^{FW}\partial_t \epsilon_{ij}^{BW}] \quad \text{(sum over } i, j\text{)}$$

General anisotropic case:

$$g_{c_{ijkl}}(\mathbf{x}) = \int_0^T dt[\epsilon_{ij}^{FW}(\mathbf{x}, t)\epsilon_{kl}^{BW}(\mathbf{x}, t)]$$

Preconditioned gradient:

$$
\begin{aligned}
B^{-1}(\mathbf{x}) &= \int_0^T dt \rho(\mathbf{x}) v_i^{FW}(\mathbf{x},t) v_i^{FW}(\mathbf{x},t) \\
\hat{g}_\eta(\mathbf{x}) &= B(\mathbf{x}) g_\eta(\mathbf{x}) \\
\eta &= \{\rho, \lambda, \mu\}
\end{aligned}
$$

Model update (steepest decent):

$$
\begin{aligned}
m_\nu^{(n+1)} &= m_\nu^{(n+1)} - \alpha_{\nu\eta} \hat{g}_\eta(\mathbf{x}) \\
\eta, \nu &= \{\rho, \lambda, \mu\}
\end{aligned}
$$

Diagonal α:

$$
\alpha = \begin{bmatrix} \alpha_\rho & & \\ & \alpha_\lambda & \\ & & \alpha_\mu \end{bmatrix}
$$

3 additional forward simulations needed to compute 3 step lengths

How to compute the gradients

Measured data:

$$
\begin{aligned}
P(\mathbf{x}_r, t) &= -\frac{1}{3}\sigma_{ii}(\mathbf{x}_r, t) \text{ (marine surface seismics)} \\
v_i(\mathbf{x}_r, t) &= \text{ (VSP, OBS)}
\end{aligned}
$$

Fields needed for the gradients:

$$
v_i^{FW}(\mathbf{x}, t),\ \epsilon_{ij}^{FW}(\mathbf{x}, t)
$$

$$
v_i^{BW}(\mathbf{x}, t),\ \epsilon_{ij}^{Bw}(\mathbf{x}, t)
$$

Elastic Kirchhoff integral

Wave equation for displacement:

$$\begin{aligned} \rho \partial_t^2 u_i &= \partial_j c_{ijkl} \partial_k u_l + f_i \\ \partial_k u_l &= \epsilon_{kl} = \text{strain} \end{aligned}$$

Kirchhoff integral:

$$\begin{aligned} u_i(\mathbf{x}, t) = & \int_0^t dt' \int_V d^3x' \, G_{ij}(\mathbf{x}, t, \mathbf{x}', t') f_j(\mathbf{x}', t') \\ & + \int_0^t dt' \int_V d^2R \, [G_{ij}(\mathbf{x}, t, \mathbf{x}_r, t') \sigma_{jk}(\mathbf{x}_r, t') n_k \\ & - \{\partial_l G_{ik}(\mathbf{x}, t, \mathbf{x}_r, t')\} c_{klmn}(\mathbf{x}_r) u_n(\mathbf{x}_r, t') n_m] \end{aligned}$$

Body force

Traction BC

Displacement BC

G_{ij} = i-component of displacement due to unit force in j-direction; Green's tensor

Sources and boundaries conditions

Monopole and dipole sources:

$$
\begin{aligned}
f_i(\mathbf{x},t) &= f_i^M(\mathbf{x},t) + f_i^D(\mathbf{x},t) \\
f_i^M(\mathbf{x},t) &= \partial_i \delta(\mathbf{x}-\mathbf{x}_s)\, M(t) \\
f_i^D(\mathbf{x},t) &= e_i \delta(\mathbf{x}-\mathbf{x}_s)\, D(t)
\end{aligned}
$$

Monopole source

Dipole source

Define

$$
\begin{aligned}
\phi_i^M(\mathbf{x},t) &= \int dR\, \{\partial_j \delta(\mathbf{x}-\mathbf{x}_r)\} c_{ijkl}(\mathbf{x}_r) u_k(\mathbf{x}_r,t) n_l \\
\phi_i^D(\mathbf{x},t) &= \int dR\, \delta(\mathbf{x}-\mathbf{x}_r)\, \sigma_{ij}(\mathbf{x}_r,t) n_j
\end{aligned}
$$

Then the Kirchhoff integral can be written as

$$
\begin{aligned}
u_i(\mathbf{x}, t) \quad &= \quad \int_0^t dt' \int d^3x' G_{ij}(\mathbf{x}, t, \mathbf{x}', t') \\
&\times [f_j^M(\mathbf{x}', t) + f_j^D(\mathbf{x}', t) + \phi_j^M(\mathbf{x}', t) + \phi_j^D(\mathbf{x}', t)]
\end{aligned}
$$

- ϕ_j^M term: Displacement/velocity data are equivalent to a monopole surface source
- ϕ_j^D term: Traction/stress/pressure data are equivalent to a dipole surface source

Elastic gradient computation

1. Forward modeling (monopole):

$$\begin{aligned} u_i^{FW}(\mathbf{x},t) &= \int_0^t dt' \int d^3x' G_{ij}(\mathbf{x},t,\mathbf{x}',t') f_j^M(\mathbf{x},t) \\ v_j^{FW}(\mathbf{x},t) &= \partial_t u_j^{FW}(\mathbf{x},t) \\ \epsilon_{ij}^{FW}(\mathbf{x},t) &= \partial_i u_j^{FW}(\mathbf{x},t) \end{aligned}$$

Compute from snapshots

2. Backward (reverse time) modeling (stress/pressure data):

$$u_i^{BW}(\mathbf{x},t) = \int_T^t dt' \int d^3x' G_{ij}(\mathbf{x},t,\mathbf{x}',t') \phi_j^D(\mathbf{x},t)$$

with $\sigma_{ij} = \sigma_{ij}^{OBS} - \sigma_{ij}^{FW}$ in ϕ_j^D

3. Compute gradients, preconditioning, steplengths as above

Gradient computation by correlation of snapshots

Forward snapshots
t_1
t_2
t_3
$\dots$
$\dots$
t_{n-2}
t_{n-1}
t_n

Backward snapshots
t_n
t_{n-1}
t_{n-2}
$\dots$
$\dots$
t_3
t_2
t_1

Elastic FWI/RTM of walkaway VSP data

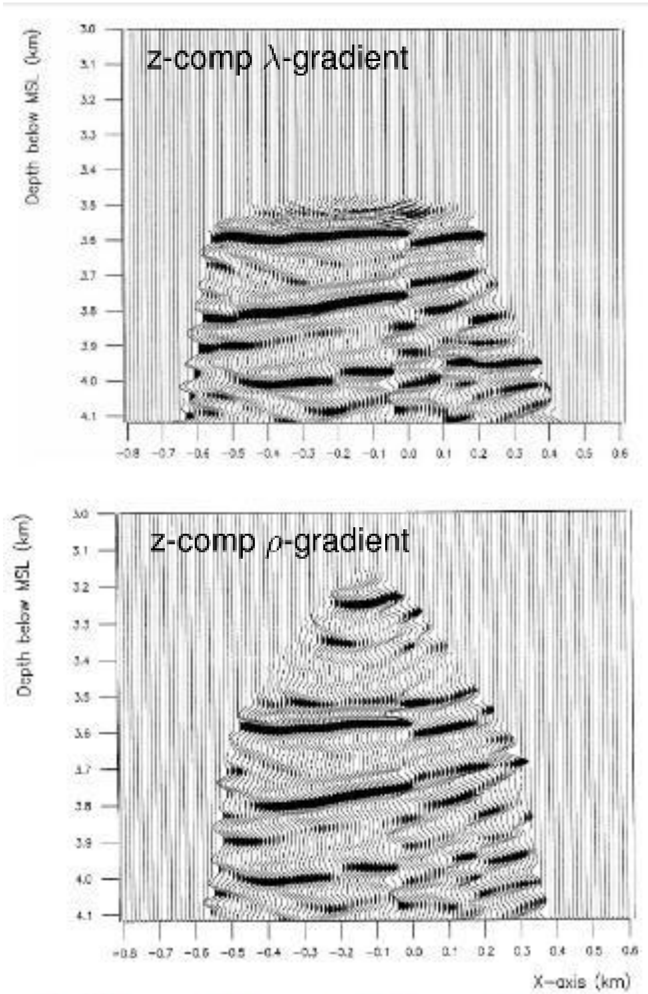


FIG. 4. Elastic reverse time migration of nine z-components for (a) P-image and (b) $P\&S$-image.

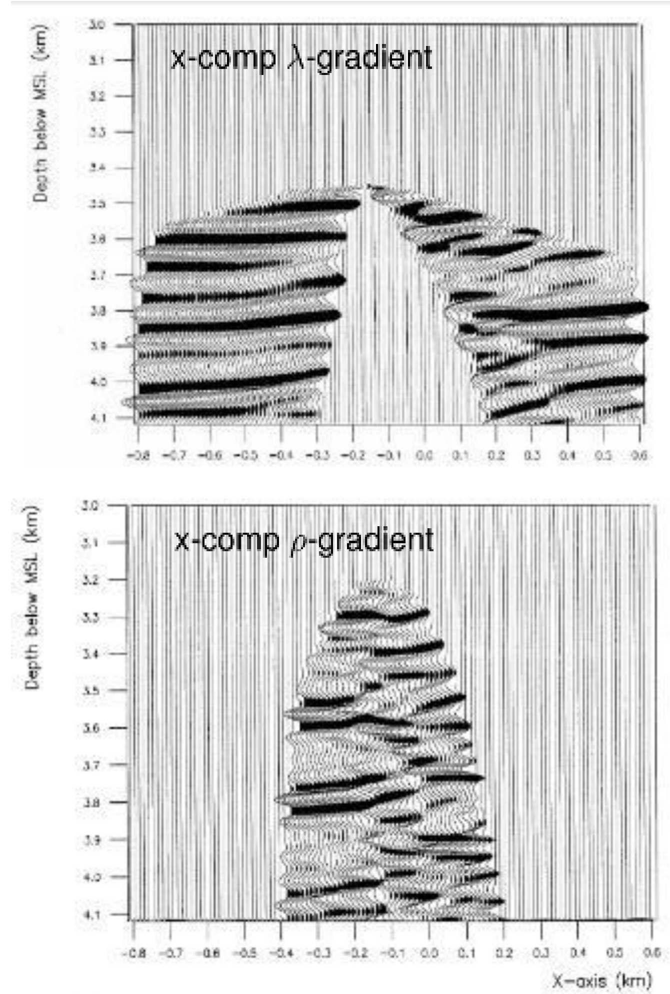
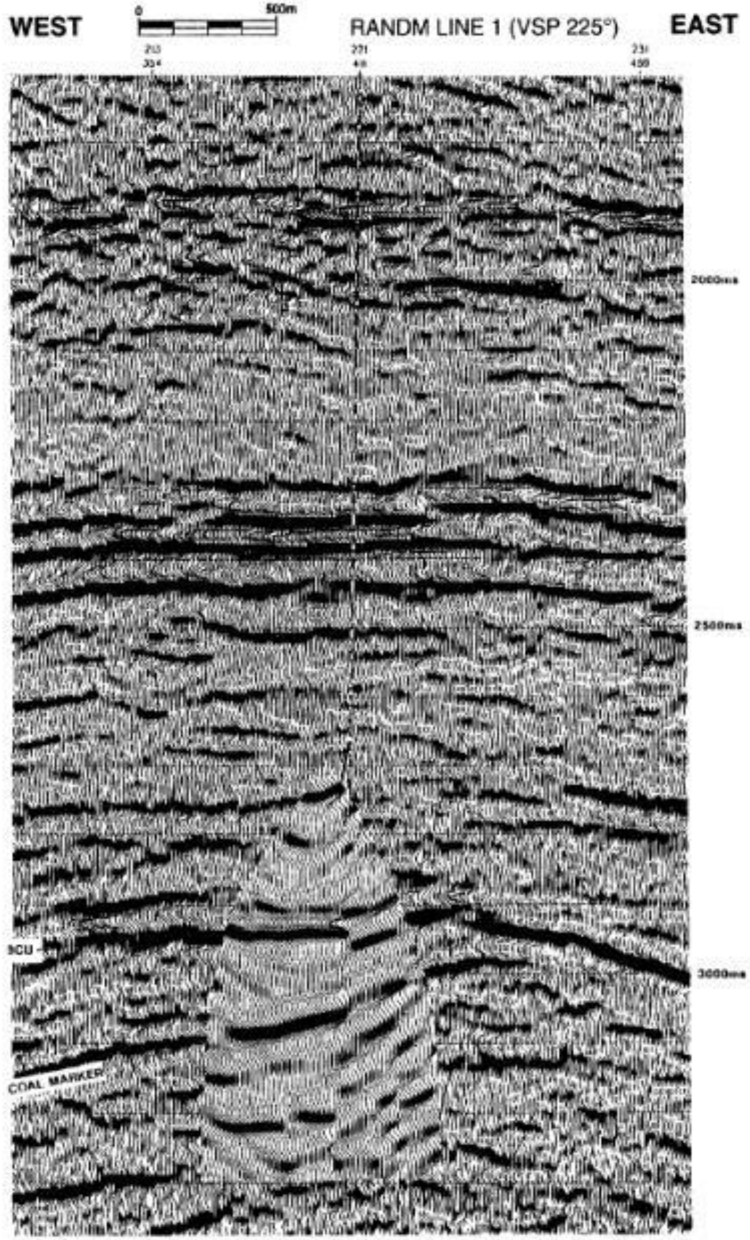
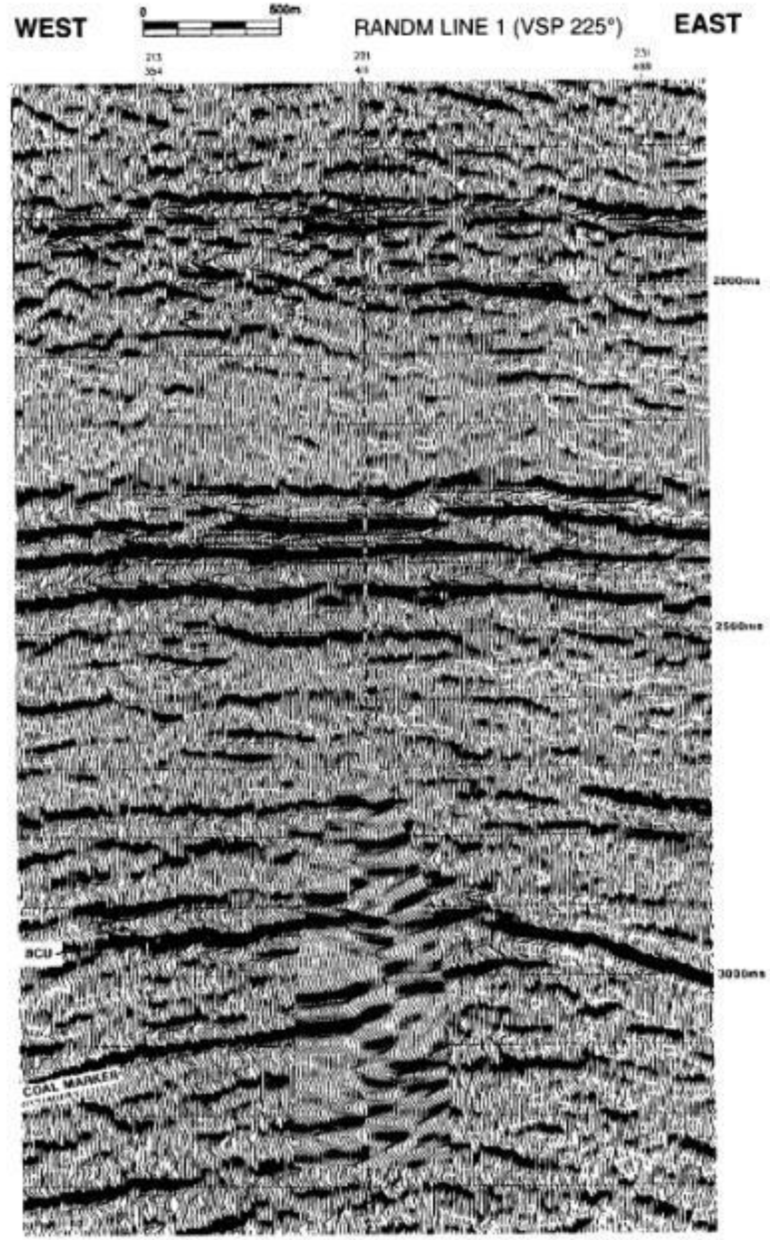


FIG. 5. Elastic reverse time migration of nine x-components. (Top) P-image; (bottom) $P\&S$-image.

Hokstad, Mittet & Landrø (1998)



z-comp ρ-gradient



x-comp ρ-gradient

Approximations to Zoeppritz equations

Aki&Richards approximation:

$$R_{PP} = \frac{1}{2}[1+\tan^2\theta]\frac{\Delta\alpha}{\alpha} - 4\left(\frac{\beta}{\alpha}\right)^2 \sin^2\theta\frac{\Delta\beta}{\beta} + \frac{1}{2}\left[1-4\left(\frac{\beta}{\alpha}\right)^2\sin^2\theta\right]\frac{\Delta\rho}{\rho}$$

Shuey approximation:

$$R_{PP}(\theta) = R_0 + G\sin^2\theta + H(\tan^2\theta - \sin^2\theta)$$

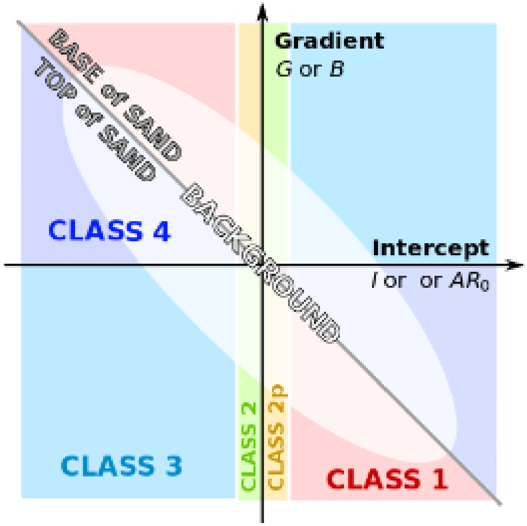
$$\begin{aligned}
R_0 &= \frac{1}{2}\left[\frac{\Delta\alpha}{\alpha} + \frac{\Delta\rho}{\rho}\right] \\
G &= \frac{1}{2}\left[\frac{\Delta\alpha}{\alpha} - 4\left(\frac{\beta}{\alpha}\right)^2\left(\frac{\Delta\rho}{\rho} + 2\frac{\Delta\beta}{\beta}\right)\right] \\
H &= \frac{1}{2}\frac{\Delta\alpha}{\alpha}
\end{aligned}$$

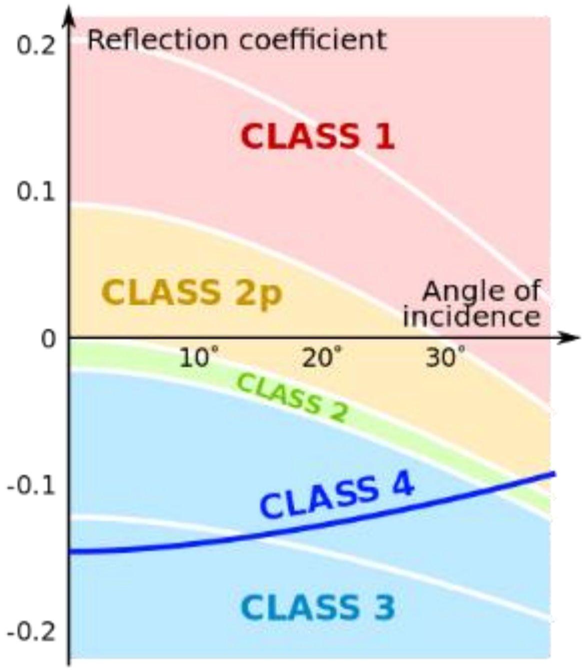
Acoustic vs elastic FWI

- Acoustic FWI (mostly) is commonly used in the seismic industry!
- Why does it work when it shouldn't?
- Reflection coefficient (gradient) depend on density, P- and S-wave velocity

$$R \simeq R_0 + G \sin^2 \theta$$

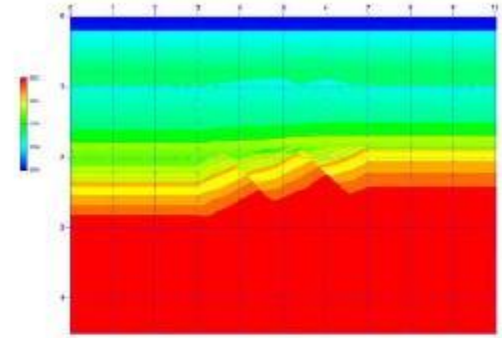
- Seismic events that can be handled by acoustic schemes:
  - ➢ Turning waves
  - ➢ Refractions (head waves)
- Reflections require elastic FWI schemes
- A possible trick: Scale out the amplitudes

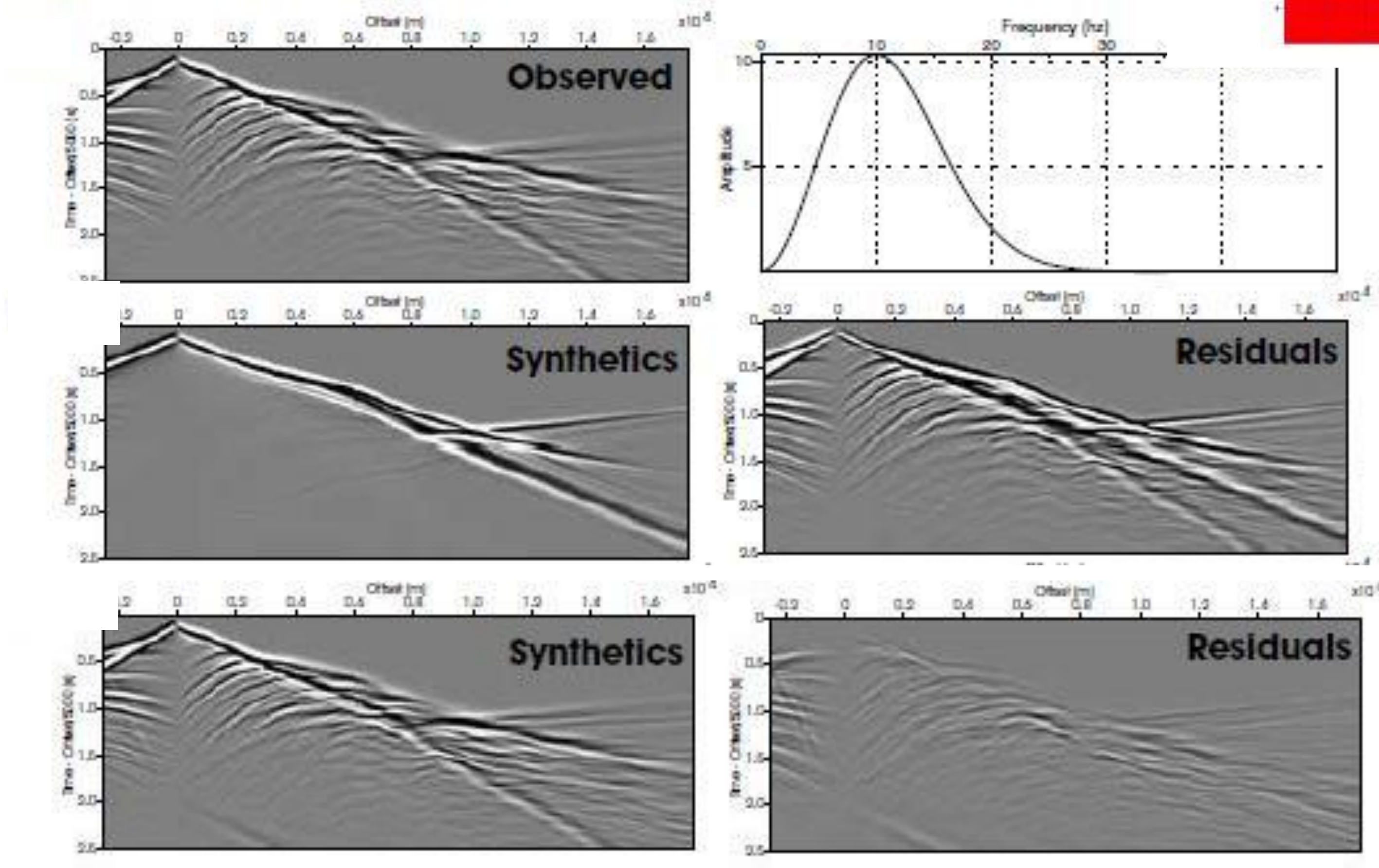




(from subsurfwiki.org)

Full waveform inversion: Data explanation



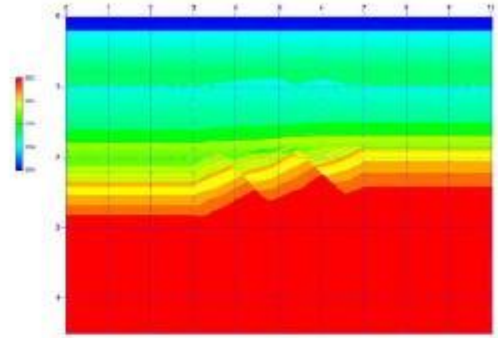


Iteration 1

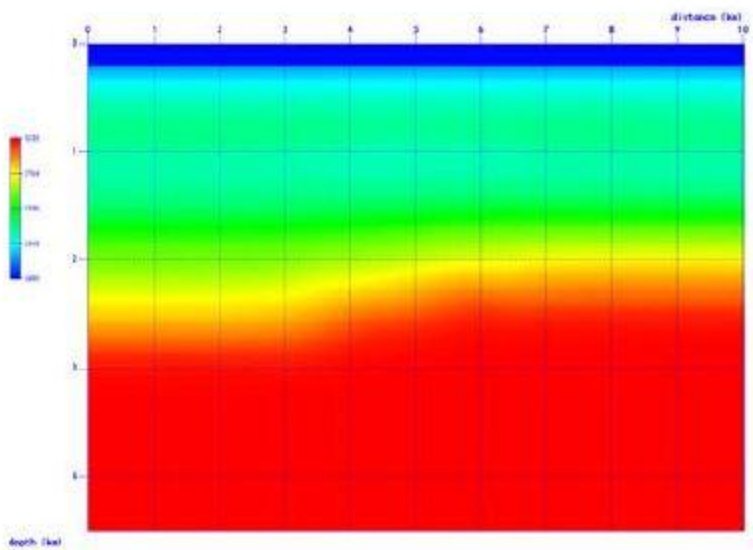
Iteration n

From Stig-Kyrre Foss (Statoil)

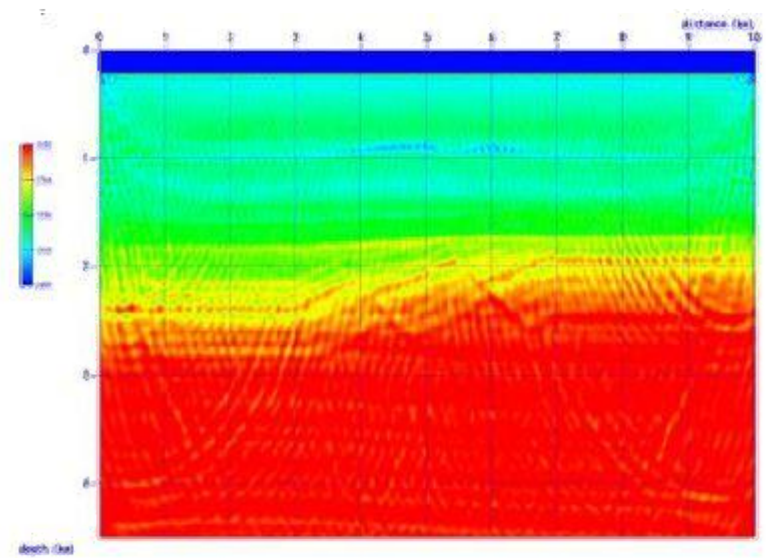
Acoustic FWI



True velocity model



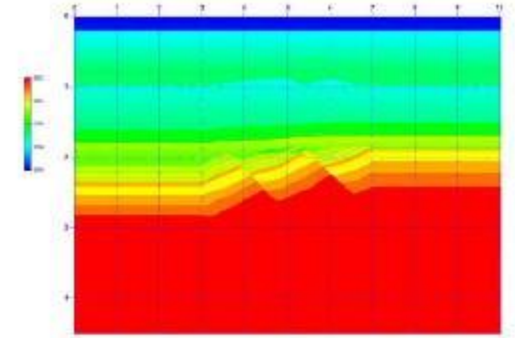
Initial model



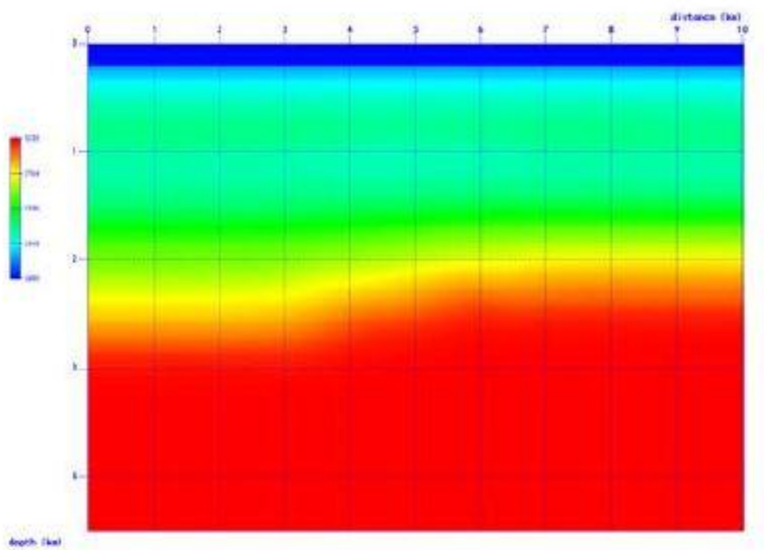
5Hz – 0-3km offset

From Stig-Kyrre Foss (Statoil)

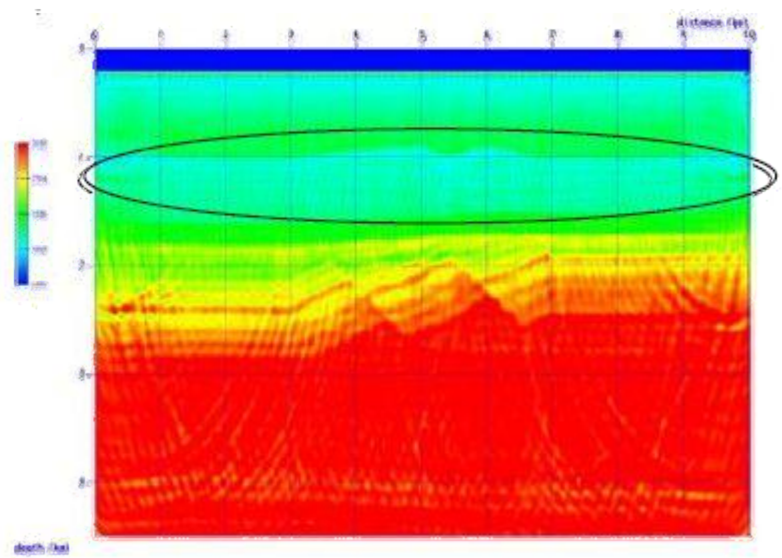
Acoustic FWI



True velocity model



Initial model

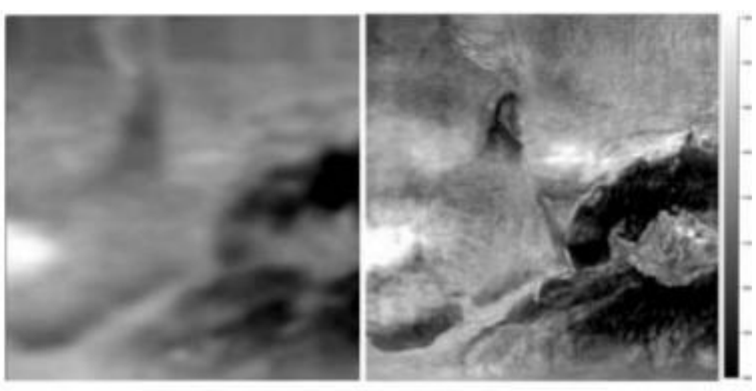


1Hz – 0-3km offset

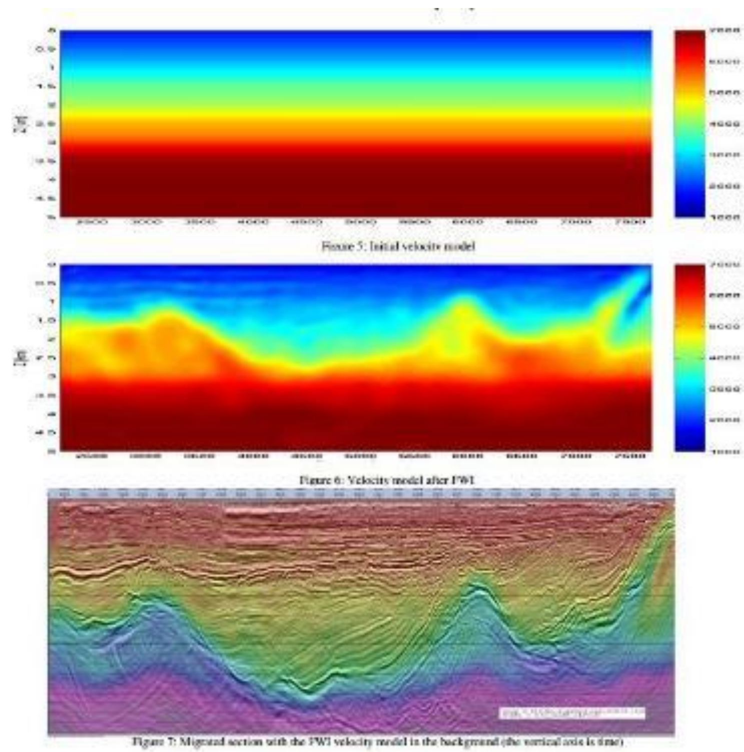
From Stig-Kyrre Foss (Statoil)

Industry examples (1)

- Full-waveform inversion has become popular in the oil industry
  - 3D acoustic FWI
  - Anisotropic modeling
  - VP0 update
  - Low frequency data

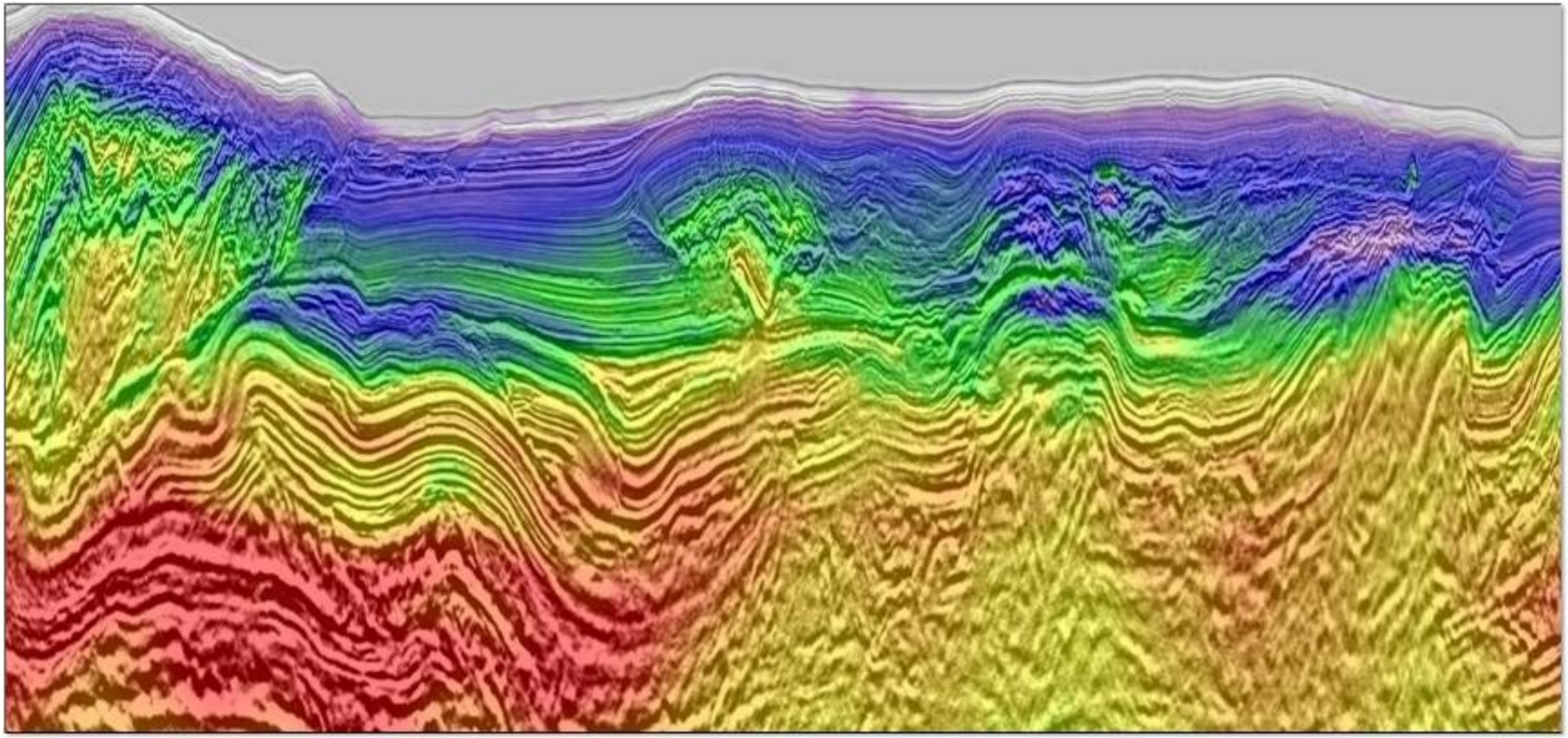


TOTAL



Shell

Industry examples (2)



Least-squares migration

Seismic inversion in general deals with two problems:

- Tomography (low frequencies)
- Imaging (high frequencies)

GEOPHYSICS, VOL. 54, NO. 12 (DECEMBER 1989); P. 1575–1586, 12 FIGS.

Inversion = migration + tomography

Peter Mora*

Least-squares migration (LSM):

- Migration within an linear-inversion framework

LSM objective function

$$\lambda = \frac{\sigma_n^2}{\sigma_m^2}$$

$$\Phi(\mathbf{m}) = |\mathbf{d} - \mathbf{L}\mathbf{m}|^2 + \lambda|\mathbf{m}|^2$$

Least-squares solution

$$\mathbf{m} = [\mathbf{L}^\mathbf{T}\mathbf{L} + \lambda\mathbf{I}]^{-1}\mathbf{L}^\mathbf{T}\mathbf{d}$$

Migration (any type in principle)

$$\tilde{\mathbf{m}} = \mathbf{L}^\mathbf{T}\mathbf{d}$$

Modeling by demigration

$$\mathbf{d} = \mathbf{L}\mathbf{m}$$

Two flavors of least-squares migration

Image-domain LSM

- The Hessian is a blurring operator

$$\tilde{\mathbf{m}} = \mathbf{L}^{\mathbf{T}}\mathbf{d}$$

$$\begin{aligned} \tilde{\mathbf{m}} &= \mathbf{H}\mathbf{m} \\ \mathbf{H} &= \mathbf{L}^{\mathbf{T}}\mathbf{L} + \lambda \mathbf{I} \end{aligned}$$

- Deblurring of the image

$$\mathbf{m} = \mathbf{H}^{-1}\tilde{\mathbf{m}}$$

- The simplest deblurring operator is the sinc function
- Point-spread functions

Data-domain LSM

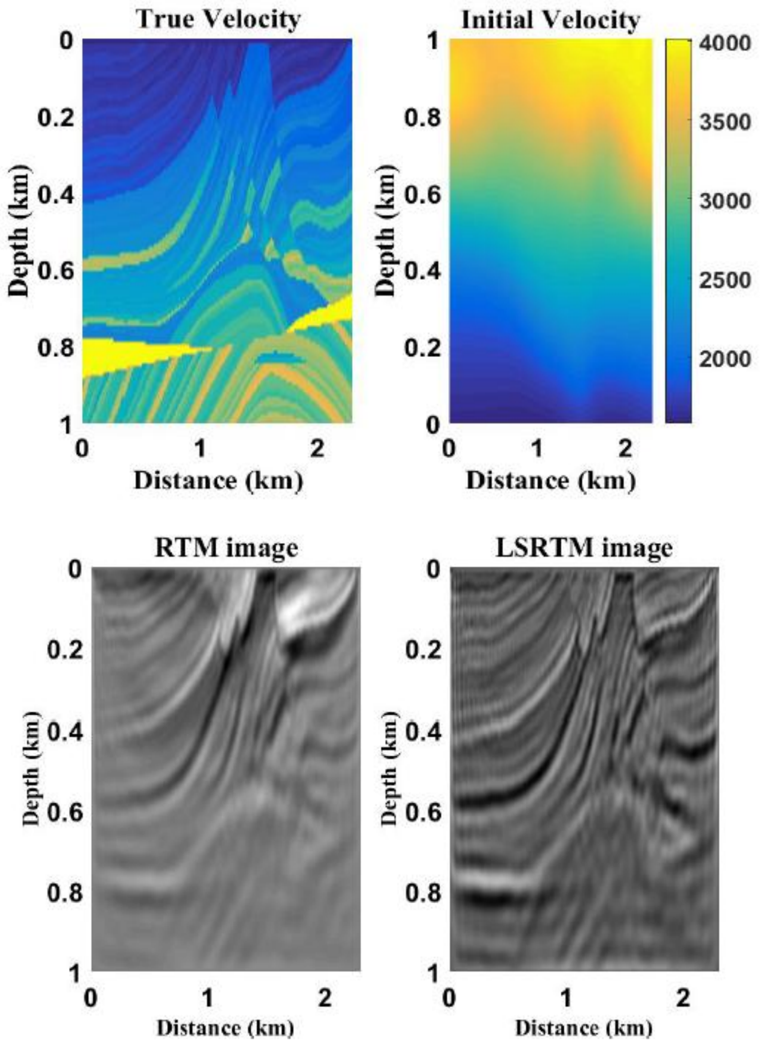
- Iterative solution
- Residual migration

$$\begin{aligned} \mathbf{m}^{(k+1)} &= \mathbf{m}^{(k)} + \mathbf{H}^{-1}\left[\mathbf{L}^{T}\mathbf{r}^{(k)}\right] \\ \mathbf{r}^{(k)} &= \mathbf{d} - \mathbf{L}\mathbf{m}^{(k)} \end{aligned}$$

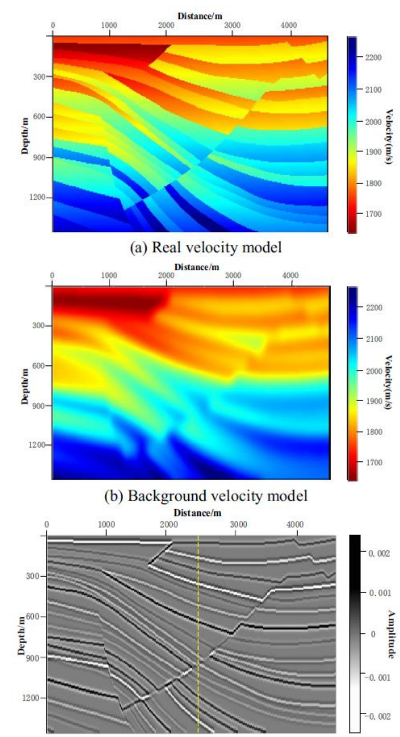
Residual migration

- The conjugate-gradient approximation to the inverse Hessian is often used

Examples: Data-domain LSM



CREWES, Yang et al. (2017)

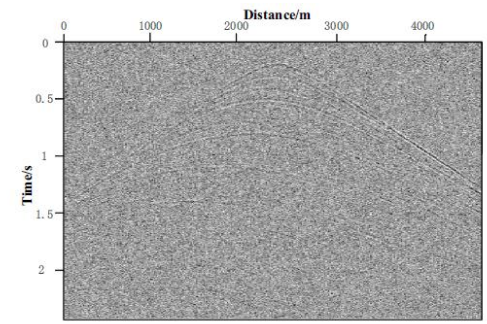


(a) Real velocity model

(b) Background velocity model

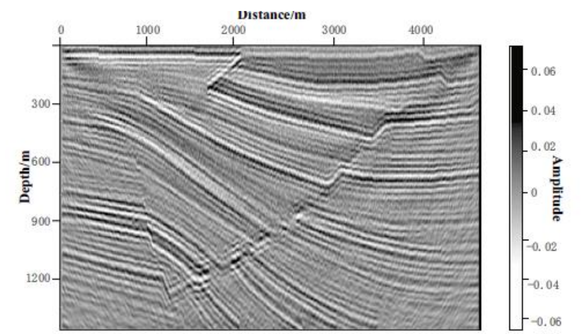
(c) Real reflection coefficient model

Figure 3. Sigsbee2B truncation velocity model.



(b) Low signal-to-noise ratio data

Figure 7. The 30th shot data without and with noise.



(b) IPRPCG-IP-LSRTM imaging result with 30 iterations

Figure 8. Low signal-to-noise ratio data imaging results of Sigsbee2

Zhang et al. (2023)

Matrix-vector formulation of adjoint state (applicable to both FWI and EM)

Time domain

$$\mathbf{M}\frac{\partial^2 \mathbf{u}}{\partial t^2} = \mathbf{A}\mathbf{u} + \mathbf{s}$$

$\mathbf{M}$ and $\mathbf{A}$ are matrices, $\mathbf{u}$ and $\mathbf{s}$ are vectors,

Wave equation for displacement:

$$\begin{aligned} \rho \partial_t^2 u_i &= \partial_j c_{ijkl} \partial_k u_l + f_i \\ \partial_k u_l &= \epsilon_{kl} = \text{strain} \end{aligned}$$

Frequency domain:

$$\begin{aligned} \mathbf{B}\mathbf{u} &= \mathbf{s} \\ \mathbf{B} &= \omega^2 \mathbf{M} - \mathbf{A} \end{aligned}$$

(Model parameters $\mathbf{m}$ are hidden inside $\mathbf{B}$)

Objective function:

$$\begin{aligned} \Phi &= \frac{1}{2}\Delta\mathbf{d}^H \Delta\mathbf{d} \\ \Delta\mathbf{d} &= \mathbf{u}_{obs} - \mathbf{u} \end{aligned}$$

(Simplified objective function; no weights, no regularization)

Matrix-form of 1st and 2nd derivatives (1D)

$$\partial_z^+ u_k \simeq \frac{u_{k+1} - u_k}{\Delta z}$$

$$\partial_z^2 u_k \simeq \frac{u_{k+1} - 2u_k + u_{k-1}}{\Delta z^2}$$

$$\partial_z \mathbf{u} \simeq -\frac{1}{\Delta z}\mathbf{Du}$$

$$\partial_z^2 \mathbf{u} \simeq -\frac{1}{\Delta z^2}\mathbf{D^T Du}$$

$$\mathbf{D} = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & 1 & -1 & \\ & & & 1 & -1 \\ & & & & 1 \end{bmatrix} \qquad \mathbf{D}^T\mathbf{D} = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}$$

in one spatial dimension

Some initial manipulations:

$$\begin{aligned} \mathbf{Bu} &= \mathbf{s} \\ \frac{\partial}{\partial \mathbf{m}}(\mathbf{Bu}) &= 0 \qquad (\mathbf{s} \text{ is independent of } \mathbf{m}) \end{aligned}$$

By the product rule

$$\mathbf{B}\frac{\partial \mathbf{u}}{\partial \mathbf{m}} = -\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\mathbf{u}$$

Define

$$\hat{\mathbf{s}} = -\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\mathbf{u} \quad \text{and} \quad \hat{\mathbf{u}} = \frac{\partial \mathbf{u}}{\partial \mathbf{m}}$$

$\hat{\mathbf{u}} = \mathbf{J} =$ Jacobian matrix

$\partial \mathbf{B}/\partial \mathbf{m}$ can be computed analytically, and is sparse

Then

$$\mathbf{B}\hat{\mathbf{u}} = \hat{\mathbf{s}}$$

Gradient:

$$\mathbf{g} = \frac{\partial \Phi}{\partial \mathbf{m}} = -\left(\frac{\partial \mathbf{u}}{\partial \mathbf{m}}\right)^H \Delta \mathbf{d} = -\mathbf{J}^H \Delta \mathbf{d}$$

Some algebra:

$$\begin{aligned}
\frac{\partial \mathbf{u}}{\partial \mathbf{m}} &= -\mathbf{B}^{-1}\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\mathbf{u} \\
\left(\frac{\partial \mathbf{u}}{\partial \mathbf{m}}\right)^H &= -\left(\mathbf{B}^{-1}\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\mathbf{u}\right)^H = -\mathbf{u}^H\left(\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\right)^H \mathbf{B}^{-H}
\end{aligned}$$

Substitute into gradient equation:

$$\mathbf{g} = \mathbf{u}^H\left(\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\right)^H \mathbf{B}^{-H}\Delta \mathbf{d} = \underbrace{\mathbf{B}^{-1}\mathbf{s}}_{\text{Forward modeling}}\left(\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\right)^H \underbrace{\mathbf{B}^{-H}\Delta \mathbf{d}}_{\text{Backward modeling}}$$

Gauss-Newton – normal equations

$$\hat{\mathbf{u}} = \mathbf{B}^{-1}\hat{\mathbf{s}} = \mathbf{J}$$

From gradient equation in previous slide, we can identify

$$\mathbf{J}^H = \mathbf{B}^{-1}\mathbf{s}\left(\frac{\partial \mathbf{B}}{\partial \mathbf{m}}\right)^H \mathbf{B}^{-H}$$

Gauss-Newton approximation:

$$\begin{aligned} \mathbf{H}\Delta\mathbf{m} &= -\mathbf{g} \\ \mathbf{H} &\simeq \mathbf{J}^H\mathbf{J} \end{aligned}$$

Model update by iterative solution to the normal equations

$$\mathbf{J}^H\mathbf{J}\Delta\mathbf{m} = \mathbf{J}^H\Delta\mathbf{d}$$

$$\mathbf{m}^{(n+1)} = \mathbf{m}^{(n)} + \Delta\mathbf{m}$$

Model updata if $\mathbf{J}^H\mathbf{J}$ or $\mathbf{J}\mathbf{J}^H$ can be inverted

$$\Delta\mathbf{m} = (\mathbf{J}^H\mathbf{J})^{-1}\mathbf{J}^H\Delta\mathbf{d} \quad \text{(least squares)}$$
$$\Delta\mathbf{m} = \mathbf{J}^H(\mathbf{J}\mathbf{J}^H)^{-1}\Delta\mathbf{d} \quad \text{(minimum norm)}$$

Moore-Penrose pseudo inverse

$$\mathbf{m}^{(n+1)} = \mathbf{m}^{(n)} + \Delta\mathbf{m}$$

The linear inverse problem revisited

$$\mathcal{F}(\mathbf{m}) = \mathbf{A}\mathbf{m}$$
$$\frac{\partial\mathcal{F}}{\partial\mathbf{m}} = \mathbf{J} = \mathbf{A}$$

$$\mathbf{m} = (\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H\mathbf{d} \quad \text{(least squares)}$$
$$\mathbf{m} = \mathbf{A}^H(\mathbf{A}\mathbf{A}^H)^{-1}\mathbf{d} \quad \text{(minimum norm)}$$

Direct solution (as seen before)

Regularization and data weights

Objective function:

$$
\begin{aligned}
\Phi &= \frac{1}{2}\Delta\mathbf{d}^H \mathbf{\Sigma}_n^{-1} \Delta\mathbf{d} + C(\mathbf{m}) \\
\Delta\mathbf{d} &= \mathbf{u}_{obs} - \mathbf{u} \\
C(\mathbf{m}) &= \text{regularization/prior}
\end{aligned}
$$

Data weights:

$$
\begin{aligned}
\mathbf{\Sigma}_n^{-1} &= \frac{1}{\sigma_d^2}\mathbf{W}^T\mathbf{W} \quad \text{(normalized weights)} \\
\Phi &= \frac{1}{2\sigma_d^2}(\mathbf{W}\Delta\mathbf{d})^H(\mathbf{W}\Delta\mathbf{d}) + C(\mathbf{m})
\end{aligned}
$$

Possible choices of data weights

Taper function depending on offset:

$$\mathbf{W} \quad \propto \quad f(\mathbf{x}_r - \mathbf{x}_s)$$

Scaling by data (independent of model):

$$\mathbf{W} \quad \propto \quad \frac{1}{|\mathbf{u}_{obs}|}$$

- This will scale out the magnitude (AVO)
- but keeps the phase information
- Ray approximation: $\phi = \omega t$
- FWI accounts for $\phi \neq \omega t$

Regularization - priors

Closeness to prior

$$C(\mathbf{m}) \quad = \quad \frac{1}{2}(\mathbf{m}-\mathbf{m}_0)^T \mathbf{\Sigma}_m^{-1}(\mathbf{m}-\mathbf{m}_0)$$

Assuming simple covariance

$$\mathbf{\Sigma}_m^{-1} \quad = \quad \frac{1}{\sigma_m^2}\mathbf{I}$$

$$C(\mathbf{m}) \quad = \quad \frac{1}{2\sigma_m^2}(\mathbf{m}-\mathbf{m}_0)^T(\mathbf{m}-\mathbf{m}_0)$$

Regularization - smoothness

Smooth solution

$$\begin{aligned} \boldsymbol{\Sigma}_m^{-1} &= \lambda \mathbf{D}^T \mathbf{D} \\ C &= \frac{\lambda}{2} \mathbf{m}^T \mathbf{D}^T \mathbf{D} \mathbf{m} = \frac{\lambda}{2} (\mathbf{D}\mathbf{m})^T (\mathbf{D}\mathbf{m}) \end{aligned}$$

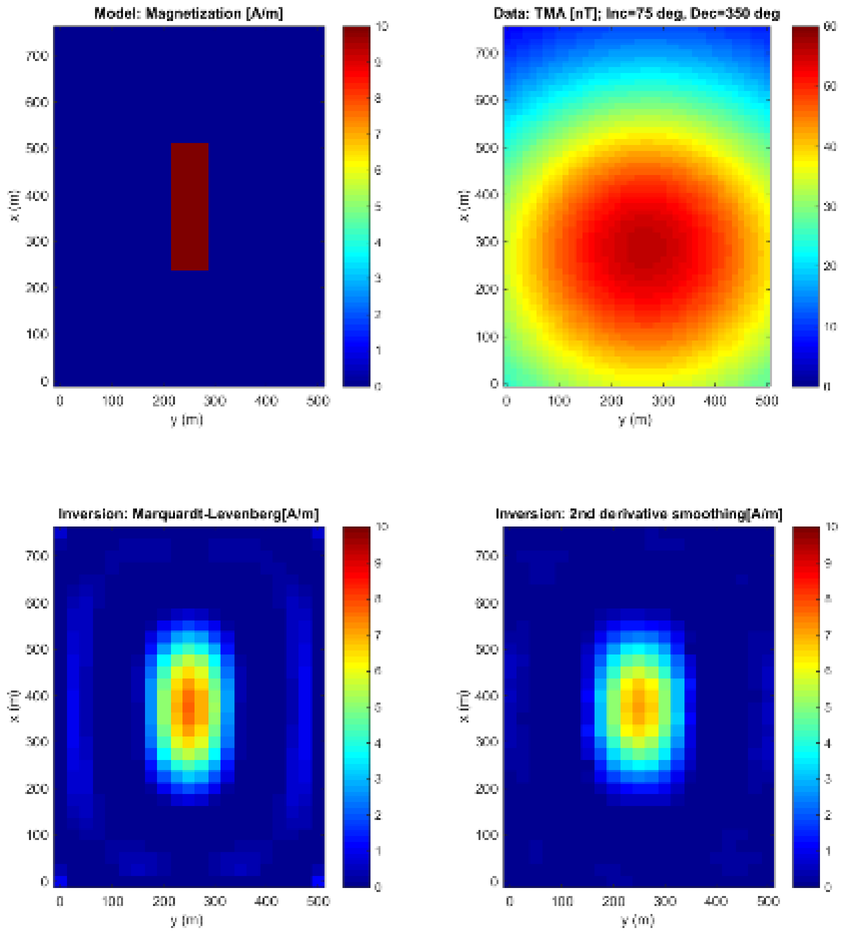
Matrix representations of 1st and 2nd derivative:

$$\mathbf{D} = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & 1 & -1 & \\ & & & 1 & -1 \\ & & & & 1 \end{bmatrix} \qquad \mathbf{D}^T\mathbf{D} = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}$$

in one spatial dimension

Sparseness: Cauchy prior (3D SRME, HiRes Radon)

Marquardt-Levenberg vs 2nd derivative smoothing



2D magnetic inversion

- Marquard-Levenberg prior (left)
- 2nd-derivative smoothing (right)